

Kenar Narayaka

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EDUCATION

University of Oxford – Christ Church

Oxford, England

BA in Computer Science & Philosophy

2025 – 2028

Relevant Coursework: Probability, Linear Algebra, Discrete Maths, Design & Analysis of Algorithms, Algorithms & Data Structures, Continuous Maths, Functional Programming, Imperative Programming, Logic & Proof, Models of Computation

Loughborough College

Loughborough, England

A-Level Maths, Further Maths, Computer Science - A*AA

2023 – 2025

EXPERIENCE

Hitachi Astemo

Jun. 2024 - Aug. 2024

Backend Engineer Intern – Human Resources

- Replaced paper-based HR processes across ~15,000 employee records with normalised database schemas and admin tooling
- Designed Dockerised REST APIs and a multi-stage approval engine with RBAC, cut request turnaround went from 7 days to 5 hours. (Postgres)
- Implemented permission-scoped CRUD dashboards and coordinate-based PDF generation wired to network printers (F3, PHP)

PROJECTS

3D Fractal Simulator | *JavaScript, three.js, GLSL*

- Built a real-time browser renderer for fractal families using signed-distance estimation and raymarching
- Implemented distance-adaptive step sizing to cut redundant samples, raising frame rate 40% and 20% more via cone-marching
- Profiled throughput against geometric detail across camera and fractal settings; scored 69/70, top OCR NEA in school history

Basketball Jumpshot Analyser | *Rust, ONNX Runtime, FFmpeg*

- Built a local video-analysis prototype running top-down RTMPose inference through ONNX Runtime over variable-frame-rate footage
- Computed six confidence-aware mechanics and timing metrics on top of cross-rep tempo consistency
- Implemented PTS-aware filtering and pose-only segmentation of continuous recordings into multiple shot reps and movement phases

Personal Site & Notes Pipeline | *JavaScript, PowerShell, Quartz, GitHub Actions*

- Built a dependency-free static site plus a pipeline publishing selected Obsidian notes to a companion subpath
- Gated publication on note frontmatter at copy time, with CI guards failing the build if an unpublished file leaks
- Hand-rolled a force-directed graph of the vault's link structure, fed by a post-build index 80% smaller

Reverse Tunnelling Reverse Proxy Minecraft Beta | *Rust, Java, Tokio, Netty, QUIC*

- Hosted single-player worlds as temporary dedicated servers over LAN, router mapping, or a QUIC relay
- Built a Rust QUIC relay and Java tunnel with session ports, reconnects, quotas, and TLS trust
- Added atomic backups, verified server downloads, and crash-safe process supervision

LEADERSHIP / EXTRACURRICULAR

Christ Church Computer Science Society

June 2026 – Present

Treasurer

- Organised 8 events including a 40-person hackathon and termly formals, planning and costing each within a £1,000 budget

Oxford Indonesia Society

June 2026 – Present

President

- Led committee planning and delivery of termly dinners and cultural events, coordinating venues, budgets, publicity, and logistics
- Revived social media and member outreach, rebuilding engagement across the Indonesian student community and

CapitOx

Jan. 2026 - Apr. 2026

Quant Academy – Oxford-Man Institute

- Built a Monte Carlo pricing pipeline for a European put under a 2-component generalised-normal mixture model (Python, NumPy)
- Implemented 3 estimators with tuned proposals, cutting variance 5x and runtime 60x vs. random-walk Metropolis-Hastings (C++, Matplotlib)

TECHNICAL SKILLS

Languages: Haskell, Scala, Python, C++, PHP, JavaScript, SQL, GLSL, Rust

Frameworks & Libraries: NumPy, SciPy, Matplotlib, three.js, Fat-Free Framework, Tauri v2

Tools: Postgres, Git, Linux, tmux, Docker

Concepts: Backend Development, Database Design, Graphics Programming, REST APIs